Derivation of the two-direction superellipse from the response form

Starting point:

$$R(v) \;=\; F\left(\sum_i w_i \,|d_i^\top v|^n\right).$$

Step 1: set up the perturbation. Perturb along the unit direction

$$\hat{v}(\varphi) \;=\; \cos\varphi\, d_1 + \sin\varphi\, d_2,$$

assuming $d_1 \perp d_2$ (so $\|\hat{v}(\varphi)\| = 1$). The perturbation angle θ parameterises the norm-matched displacement on the sphere, so the perturbation is $\sin\theta \cdot \hat{v}(\varphi)$. The projections onto the two directions are

$$d_1^\top\big(\sin\theta\,\hat{v}\big) = \sin\theta\cos\varphi, \qquad d_2^\top\big(\sin\theta\,\hat{v}\big) = \sin\theta\sin\varphi.$$

For the small angles relevant to breaking, $\sin\theta \approx \theta$, so we use $\theta\cos\varphi$ and $\theta\sin\varphi$ directly.

Step 2: plug into the response. Assume the remaining features d_i ($i \neq 1, 2$) have negligible overlap with $\hat{v}(\varphi)$ — the relevant approximation when d_1, d_2 are isolated feature directions. Then the inner sum collapses to two terms:

$$R \;=\; F(w_1(\theta\cos\varphi)^n + w_2(\theta\sin\varphi)^n) \;=\; F\big(\theta^n\big[w_1\cos^n\varphi + w_2\sin^n\varphi\big]\big)\,.$$

Step 3: breaking condition. The plateau breaks when F crosses a fixed level F^*, equivalently when its argument crosses a fixed scalar ξ^*. The breaking angle $\theta(\varphi)$ satisfies

$$\theta(\varphi)^n\big[w_1\cos^n\varphi + w_2\sin^n\varphi\big] \;=\; \xi^*. \tag{1}$$

Step 4: calibrate w_1, w_2 from the two axis sweeps. Along d_1 alone ($\varphi = 0$), (1) gives $\theta_1^n\, w_1 = \xi^*$, hence

$$w_1 \;=\; \xi^*/\theta_1^n.$$

Along d_2 alone ($\varphi = \pi/2$), similarly

$$w_2 \;=\; \xi^*/\theta_2^n.$$

Step 5: substitute and simplify. Plug w_1, w_2 into (1) and divide by ξ^*:

$$\theta(\varphi)^n\left[\frac{\cos^n\varphi}{\theta_1^n} + \frac{\sin^n\varphi}{\theta_2^n}\right] \;=\; 1,$$

which rearranges to

$$\boxed{\left(\frac{\theta(\varphi)\cos\varphi}{\theta_1}\right)^n + \left(\frac{\theta(\varphi)\sin\varphi}{\theta_2}\right)^n \;=\; 1.}$$

This is a superellipse of exponent n in the normalised coordinates

$$\big(\theta(\varphi)\cos\varphi/\theta_1,\;\theta(\varphi)\sin\varphi/\theta_2\big).$$

Assumptions used.

- $d_1 \perp d_2$, so $\hat{v}(\varphi)$ is unit norm and the two projections are independent.
- Other features d_i have negligible overlap with $\hat{v}(\varphi)$ for all φ, so they drop out of the inner sum.
- Small-angle regime, $\sin\theta \approx \theta$, so the sphere parameterisation coincides with the angle used in the final expression.
- F is strictly monotonic through the threshold region, so "breaking" corresponds to a fixed value ξ^* of its argument.

The first two assumptions are what make the two-direction fit work pair-by-pair without needing the full feature catalogue: within a single pair's subspace, only w_1, w_2 appear, and both are absorbed into θ_1, θ_2. The superellipse shape is then purely a statement about n.